

The cholesterol-lowering effect of coconut flakes in humans with moderately raised serum cholesterol.

This study investigated the effect of coconut flakes on serum cholesterol levels of humans with moderately raised serum cholesterol in 21 subjects. The serum total cholesterol of subjects differed and ranged from 259 to 283 mg/dL. The study was conducted in a double-blind randomized crossover design on a 14-week period, consisting of four 2-week experimental periods, with each experimental period separated by a 2-week washout period. The test foods were as follows: corn flakes as the control food, oat bran flakes as the reference food, and corn flakes with 15% and 25% dietary fiber from coconut flakes (made from coconut flour production). Results showed a significant percent reduction in serum total and low-density lipoprotein (LDL) cholesterol (in mg/dL) for all test foods, except for corn flakes, as follows: oat bran flakes, 8.4 $\pm$ 1.4 and 8.8 $\pm$ 6.0, respectively; 15% coconut flakes, 6.9 $\pm$ 1.1 and 11.0 $\pm$ 4.0, respectively; and 25% coconut flakes, 10.8 $\pm$ 1.3 and 9.2 $\pm$ 5.4, respectively. Serum triglycerides were significantly reduced for all test foods: corn flakes, 14.5 $\pm$ 6.3%; oat bran flakes, 22.7 $\pm$ 2.9%; 15% coconut flakes, 19.3 $\pm$ 5.7%; and 25% coconut flakes, 21.8 $\pm$ 6.0%. Only 60% of the subjects were considered for serum triglycerides reduction (serum triglycerides >170 mg/dL). In conclusion, both 15% and 25% coconut flakes reduced serum total and LDL cholesterol and serum triglycerides of humans with moderately raised serum cholesterol levels. Coconut flour is a good source of both soluble and insoluble dietary fiber, and both types of fiber may have significant role in the reduction of the above lipid biomarker. To our knowledge, this is the first study conducted to show a relationship between dietary fiber from a coconut by-product and a lipid biomarker. Results from this study serves as a good basis in the development of coconut flakes/flour as a functional food, justifying the increased production of coconut and coconut by-products.